



MassMutual Institutional Solutions

Business Strategy Overview

JANUARY 2023

Strategic Summary: Institutional Solutions

January 2023

Market leadership and competitive advantages provide scalable businesses with attractive profits

Business Overview

- **Institutional Solutions** serves the financial needs of **business customers** via sustainable leadership positions across **four distinct markets**
- Institutional Solutions delivers **significant value** to policyowners and enterprise strategy through **scalable business growth** and **high earnings**
- **GAAP Pre-Tax Earnings** have grown from \$386M (2016) to **\$874M (2022)** representing a **substantial portion of MassMutual earnings and OBE's**

Markets and Competition

- Secular trends reinforce the **long-term attractiveness of the institutional markets** and the **strategic business opportunities**
 - Increasingly **institutional customer demand** across markets favors providers with **scale, expertise, financial strength and advanced analytics**
- MassMutual's **institutional presence** is consistent with key peers, and provides **broader coverage across markets** than other insurers
- Institutional Solutions has established **leadership positions** in the key institutional markets, with the goal to **optimize profit, growth and risk**

Value Proposition & Market Approach

- Institutional Solutions value proposition focuses on **financial strength, market expertise, innovation, flexibility, intermediary relationships and trust**
- A **deliberate approach to the market** enabled business success over the past six years and **guides future business strategy**
 - **Primary customer is the institution** as the purchase and retention decision resides with the business customer
 - Target market segments are those **customers who value high-quality** providers, with particular focus on **mid-to-large size institutions**
 - Distribution is focused on **deep relationships with unaffiliated intermediaries** as distribution partners are **specialized in each market**
 - **Customer experience** with tailored delivery and ease of doing business, through an infrastructure that **leverages the benefits of scale**
 - Dynamic **specialized talent** and **robust data analytics** are the foundation for all business capabilities
- Institutional Solutions **competitive advantage** differentiates on these elements and maintains focus on **target customer segments**

Risks, Constraints & Challenges

- **External risks** are factors impacting **market opportunity** such as **economic conditions, interest rates, regulatory policy and non-traditional entrants**
- **Internal factors** influencing **growth potential** include **enterprise constraints** around **capital, reinsurance leverage** and **invested assets**
- 2023 business focus is on **optimizing profitability within these constraints** while further developing **infrastructure, emerging solutions and capabilities**

Financials and Execution

- **Advancing business strategy** will leverage **competitive advantages** and **leadership positions**, along with **strong execution** and **business investment**
- Growth in **Assets Under Management** from \$52B (2016) to **\$105B (2022)** via new business and retention drives the growth in **GAAP Pre-Tax Earnings**
- **Institutional Solutions** strategy delivers **favorable return businesses** that provide **capital-efficient and early-emergent profitability** supporting GAAP **earnings**, policyowner **dividends**, **statutory** profits, **capital** generation and enterprise **strategic investments**

What is the purpose of MassMutual Institutional Solutions?

January 2023

MISSION

“Trusted to Make Institutions Stronger”

VISION

Institutions rely on MassMutual for the expertise and financial strength to help them better meet the needs of their owners, employees, customers and communities. Meeting today's business challenges requires a genuine provider that empowers institutional customers to succeed and makes companies stronger for the long-term by enabling them to focus on what they do best.

We are the premier provider of comprehensive institutional protection and investment solutions through deep experience in institutional markets, keen focus on evolving customer needs, innovative approach to solution design, and clear delivery on client experience.

ENTERPRISE PURPOSE

Institutional Solutions strategic purpose is to deliver scalable business growth and attractive earnings (OBEs) that contribute to MassMutual's enterprise goals. Business decisions focus on the value to MassMutual's strategic objectives across the company's businesses.

What are the Institutional Solutions business markets?

January 2023

Institutional Insurance (BOLI/COLI)

Delivers insurance solutions designed to meet the clients needs for key person protection, tax-efficient benefit funding and asset accumulation. Product solutions include bank owned life insurance (BOLI) and corporate owned life insurance (COLI).

Pension Risk Transfer

Serves businesses seeking to transfer pension liabilities to an insurance provider. We offer general account and separate account group annuities in the U.S. market, along with other longevity reinsurance solutions in the non-U.S. market.

Stable Value Investments

Provides principal protection solutions on an investment only basis to retirement plan advisers and defined contribution plan sponsors. These guaranteed solutions include a combination of asset management and stable value insurance wraps.

Funding Agreements

Provides institutions with solutions that offer fixed income returns for a set time period to meet investment needs. Funding agreement backed notes serve the broad fixed income market, while Muni GIC's meet the investment needs of municipalities.

Institutional Solutions aligns with MassMutual's strategic imperatives

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MassMutual Purpose

Why We Exist

Help people secure their future and protect the ones they love

MassMutual Vision

What We Want to Become

Provide financial well-being for all Americans

MassMutual Values

What We Believe

We are deeply committed to integrity, customer focus, and collaboration to achieve the best results – all while valuing each other

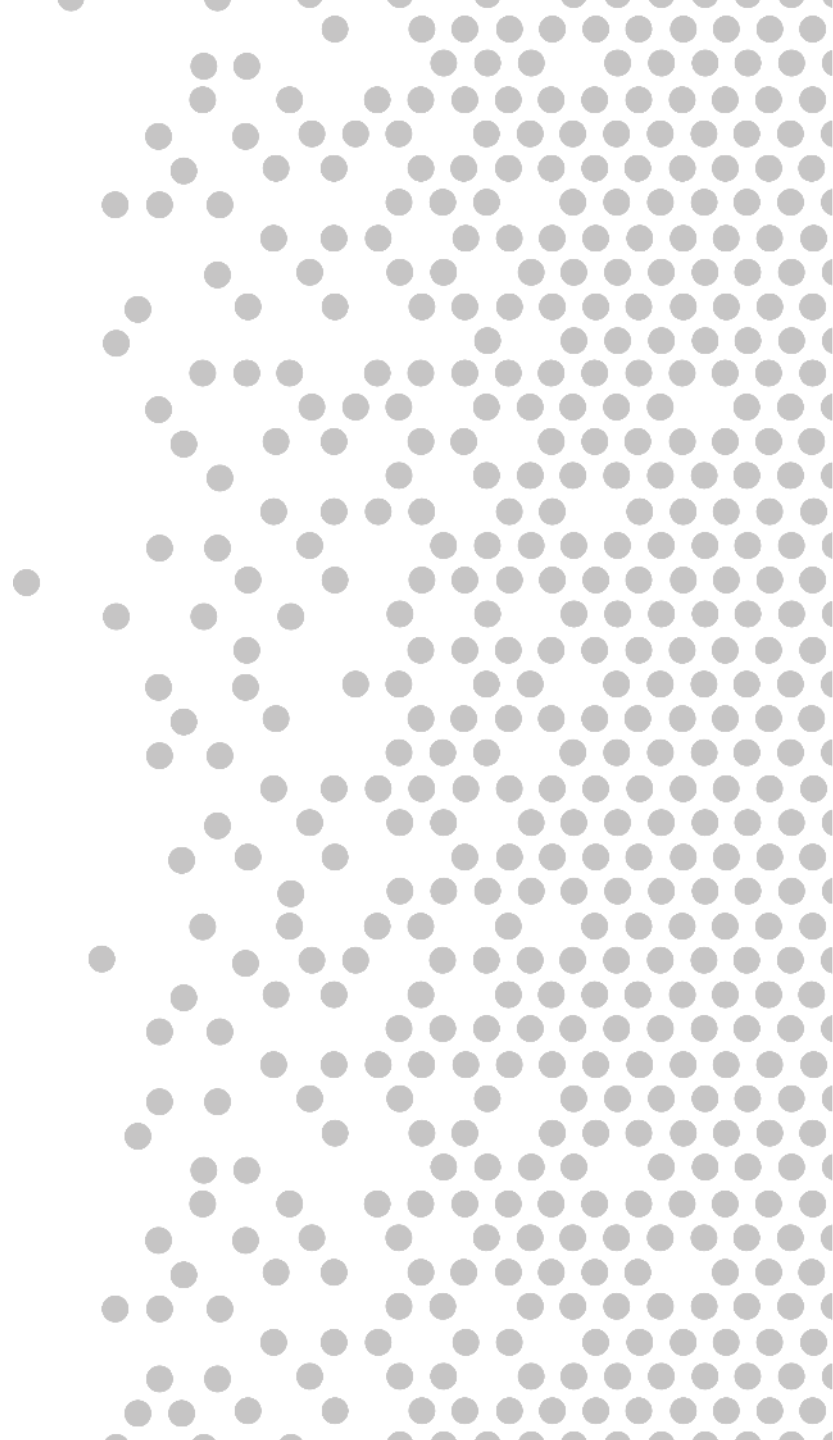
MassMutual's Strategic Areas of Focus



MassMutual 175 Aspirations		
	Earnings Growth Sustainably grow run-rate earnings by \$750M	✓✓
	Customers Grow our customers and members to 25 million	✓
	Digital Make all of our customer journeys mobile-enabled, with at least half of our interactions carried out digitally	✓
	Net Promoter Score Achieve industry-leading Net Promoter scores in each of our channels	✓

○ ✓ Institutional Solutions alignment

Institutional Solutions Business Model



While the various institutional markets are distinct, certain characteristics of MassMutual's "value proposition" resonate across markets, customers and advisors. In a space where customization and complexity are common, the key elements of MassMutual's value proposition include:

- Financial Strength as exhibited by a strong balance sheet and high credit ratings
- Innovation and Flexibility in developing solutions to meet specific customer needs (within MM risk tolerances)
- Intermediary Relationships to partner with consultants and advisors
- Market Commitment in terms of presence in the market and consistency of market position
- Expertise in the market demonstrated through providing thought leadership, knowledge and consultation to clients and advisors
- Focus to compete on opportunities with favorable prospects and where MassMutual's value proposition resonates
- Brand and Reputation such that customers are willing to pay for value
- Genuineness and Trust while delivering on our promises and working with customers to understand and meet their needs

Collectively, our portfolio of institutional businesses positions MassMutual as a comprehensive solutions provider that helps customers meet their diverse financial needs. In each market, IS focuses on the target market segments that most value these attributes.

BOLI / COLI	Pension Risk Transfer	Stable Value Investments	Funding Agreements
<u>Additionally:</u> • Serve both large and small bank clients • Responsive product • Consistency of market presence • Efficient new business process	<u>Additionally:</u> • Flexible solutions (client needs focus) • Ability to handle complexity • Key account relationships • Nexus of financial strength and price point • AIK expertise	<u>Additionally:</u> • Flexible solution design (client focus) • Stable value heritage and expertise • Key account relationships • Adjacent markets • Financial strength	<u>Additionally:</u> • Strong credit profile • Consistent market presence • High bond investor interest in our name • Company strength

CUSTOMERS: Across these businesses, the common market characteristic is an institutional business customer seeking financial solutions to address a need at the enterprise level. While individuals are part of some solutions, the driving need and ultimate decision (purchase and retention) resides with a company's need related to their balance sheet, income statement, benefit offerings or risk profile.

MARKETS: Within each solution, the market characteristics differ based on factors such as economic environment and market maturity. Whether "growth market" or "take share" or "leader position" opportunity, the focus is on profitable growth in a risk managed manner.

BOLI / COLI	Pension Risk Transfer	Stable Value Investments	Funding Agreements
<u>Customers:</u> Large and small businesses seeking an attractive investment asset and benefit funding vehicle <u>Market:</u> ~\$3-4 billion in annual BOLI sales; ~\$2 billion in annual COLI sales <u>Market Trend:</u> Low growth market (usually \$4 billion); material level of sales; market size varies with tax rates and interest rates; 2021/2022 atypically high demand given bank cash levels	<u>Customers:</u> Public and private companies seeking to transfer pension liabilities <u>Market:</u> ~\$38 billion in annual sales (excluding mega-jumbos) <u>Market Trend:</u> High growth market; further acceleration of market growth occurs with rising interest rates and tax reform	<u>Customers:</u> Sponsors and institutions seeking safe return options <u>Market:</u> SVIO wrap market at ~\$500+ billion AUM <u>Market Trend:</u> SVIO wrap market usually flat, some provider consolidation; growth when economic uncertainty	<u>Customers:</u> Institutions seeking investment grade returns; municipalities seeking investment vehicle for bond proceeds <u>Market:</u> ~\$37 billion annual FABN issuance (2022) and ~\$150 billion AUM (FABN); flat market (Muni GIC) <u>Market Trend:</u> Managed growth
Leader Position	Growth Market	Leader Position	Leader Position

The focus is on the customer need, not the product or service itself. While we offer specific solutions and products in the marketplace, each is a means to address an institution's need related to insurance, investment/accumulation, income, funding, pension, returns, mortality or longevity. New solutions development has focused on adjacent offerings that help institutional clients address their evolving needs.

Prospectively, MassMutual continues to assess institutional customer needs and preferences for solutions to meet financial business risks and goals. Where opportunities exist to achieve profitable returns within MM risk limits while leveraging MM competitive strengths, MM develops new institutional solution offerings that align with enterprise strategic goals.

BOLI / COLI	Pension Risk Transfer	Stable Value Investments	Funding Agreements
Primary Solutions <u>Today:</u> • New Money Rate BOLI (SL19) • Portfolio Rate BOLI (SL11B) • RVUL COLI (SL18) • PPVUL NQ COLI (SL20) <u>Tomorrow:</u> • PPVUL iCOLI (SL21)	Primary Solutions <u>Today:</u> • General Account Group Annuity • Commingled Separate Account Group Annuity • Annuity Buy-In • UK Longevity Reinsurance <u>Tomorrow:</u> • Expanded Reinsurance Partnerships • Non-U.S. Longevity Reinsurance	Primary Solutions <u>Today:</u> • Synthetic Wrap • Separate Account • Collective Investment Trust • Pooled Separate Account • Traditional GIC • General Account	Primary Solutions <u>Today:</u> • FABN Syndicated • FABN Reverse • FABN Non-USD • Muni GIC issuance • Muni GIC block assumption <u>Tomorrow:</u> • "Green" FABN • Commercial Paper FABN

In most of the institutional markets, the distribution partners are highly specialized given the technical nature of the solutions and the level of complexity and customization involved. Not surprisingly, a small number of advisor and consultants often control the majority of each market. Smaller players serve specific niche segments or small case opportunities. With some of the more financial market oriented solutions, the larger Wall Street and investment firms are actively involved as distribution partners.

In certain of the higher growth markets, non-traditional distribution partners occasionally emerge particularly where large players in adjacent markets try to capture additional advisory and placement revenue.

BOLI / COLI	Pension Risk Transfer	Stable Value Investments	Funding Agreements
<u>Key Distribution Partners:</u> • Lynch & Associates • Bank Fin'l Services • The Todd Org. • NFP Executive Benefits • The BOLI Group • Institutional Insurance Group • Newport Group • Lyon Street	<u>Key Distribution Partners:</u> • Willis Towers Watson • Aon Hewitt • Mercer • Milliman • BCG Pension Risk • Deloitte	<u>Key Distribution Partners:</u> • Galliard • Invesco • Vanguard • T. Rowe Price • GSAM • Morley • Federated • Putnam	<u>Key Distribution Partners:</u> • JP Morgan • Bank of America • US Bank • Morgan Stanley • Goldman Sachs • Credit Suisse • BNP Paribas • Mizuho • Deutsche Bank

The competitive landscape varies across the institutional markets in which MassMutual competes. For those solutions with more of a guarantee or actuarial element, the large established insurance providers are often our competitors. In the more asset based solutions, the large investment and financial firms commonly inhabit this space. Where markets are mature and relatively stable, new entrants are uncommon. In fact, in some cases noted below, industry consolidation is taking place. In the growth markets, new and returning competitors appear on a frequent basis. Private equity backed issuers are more common. MassMutual seeks to compete where our value proposition resonates, and where conditions favor attractive risk-adjusted returns.

BOLI / COLI	Pension Risk Transfer	Stable Value Investments	Funding Agreements
Key Competitors - Northwestern Mutual - New York Life - Protective (Great West) - Midland - Securian (MN Life) - Nationwide - Prudential - MetLife	Key Competitors - MetLife - Athene - Principal - PacLife - Prudential - Legal & General - AIG - Securian <u>Note:</u> - New providers each year	Key Competitors - Prudential - MetLife - Transamerica - PacLife - Voya - AIG <u>Note:</u> - No meaningful new entrants - Some consolidation	Key Competitors - MetLife - New York Life - Northwestern Mutual - Principal - Guardian - PacLife - Jackson - Athene - Several Others <u>Note:</u> - Muni GIC – banks

The level and volatility of market interest rates and credit spreads is the most common trend factor across the institutional markets ... both in terms of impact on market demand and effect on in-force spreads/yields. Within the financial-related trends, any changes resulting from emerging capital standards would apply to several of these solutions ... some require capital up-front while others draw capital in certain situations. Similarly, overall economic conditions, inflation rates, geopolitical concerns, the pandemic and the impacts of US fiscal/political matters effect all institutional markets.

Many of the other environmental trends are specific to each market need. The larger categories relate to regulatory developments, industry provider conditions and shifting consumer preferences. The impact of evolving customer experience (CX) expectations is addressed on a separate page.

BOLI / COLI

Most relevant trends:

- Interest rates (both levels and pace of change that can impact solution attractiveness)
- Bank capital levels and impact on demand
- Tax code reform that impacts the tax advantaged status or corporate tax rate
- Economic sentiment

Pension Risk Transfer

Most relevant trends:

- Interest rates (from both a pension valuation and debt funding basis)
- Market demand vis-à-vis insurance provider capacity
- PBGC premiums
- New entrants given growth opportunities
- Reinsurance
- Longevity trends
- Economic sentiment

Stable Value Investments

Most relevant trends:

- Institutional client preferences for multi-manager asset management
- Changes in the entrance/exit of wrap providers
- Rising interest rates (where increased rates could cause MV<BV)
- Fee pressures
- Yield seekers

Funding Agreements

Most relevant trends:

- Interest rates and yield curve (which influence market attractiveness of coupon rates)
- Volatility in debt markets that impact issuance conditions
- Market preference for strong credit names
- Geopolitics
- Economic sentiment

Institutional Solutions: Customer Experience (CX) and Operating Model

January 2023

The nature of the distribution and service model for these businesses necessitates that customer experience applies to the institution, the intermediary and the individual. Each of these customers is involved to varying degrees and at different points in the life cycle depending on the specific business. In all cases, the customer experience needs to focus on ease of doing business, customized service delivery and high quality accurate results.

The operating model and digital state is evolving across the businesses. The focus is on assuring market parity and driving differentiation in the customer experience and digital space. The goal is to enhance the customer experience in light of market needs, as well as use operating models that leverage scale, complimentary activities and the most efficient services.

BOLI / COLI

CX Importance:

- Institution – Primary
- Intermediary – Primary
- Individual – Secondary

Operations Model:

- Internal Ops Team
- Various Admin Platforms
- Leverage Imaging

Areas of Focus:

- Scale/Efficiency
- Digital Enablement

Pension Risk Transfer

CX Importance:

- Institution – Primary
- Intermediary – Primary
- Individual – Secondary

Operations Model:

- Internal Ops Team
- External Admin Platform
- Pension Annuity Website

Areas of Focus:

- Digital Enablement
- Ops Value Stream
- Scale/Speed

Stable Value Investments

CX Importance:

- Institution – Primary
- Intermediary – Primary
- Individual – Not Applicable

Operations Model:

- Internal Ops Team
- Admin Platform Conversions

Areas of Focus:

- SVIO Admin Platform
- Scale and Diversification
- Digital Enablement
- Infrastructure

Funding Agreements

CX Importance:

- Institution – Primary
- Intermediary – Primary
- Individual – Not Applicable

Operations Model:

- Internal Ops Team
- SAP Platform

Areas of Focus:

- Ops Processes
- Digital Enablement

Institutional Solutions: Financial Position Overview

January 2023

Institutional Solutions contributes a material component of MassMutual's distributable earnings and top-line business growth. Collectively, these businesses provide a very attractive profit profile and the ability to achieve results on a scale basis. Aligned directly with enterprise priorities, the institutional offerings contribute meaningfully to MassMutual's customer goals and strategic purpose.

Prospectively, Institutional Solutions seeks to continue to increase these financial and business results (within company risk/capacity limits) through delivering comprehensive solutions to institutions and helping the success of other MassMutual businesses.

BOLI / COLI

Core Pre-Tax GAAP Earnings:
~\$388 million

AUM:
\$36 billion

Sales (BOLI):
'16A – \$1.4 billion
'17A – \$1.3 billion
'18A – \$0.6 billion
'19A – \$1.3 billion
'20A – \$0.9 billion
'21A – \$2.4 billion
'22A – \$1.5 billion

Industry Rank ('22 – Est.):
#2 General Account BOLI Sales

Pension Risk Transfer

Core Pre-Tax GAAP Earnings:
~\$314 million

AUM:
\$17 billion

Sales:
'16A – \$2.0 billion
'17A – \$2.2 billion
'18A – \$1.9 billion
'19A – \$2.8 billion
'20A – \$3.1 billion
'21A – \$3.9 billion
'22A – \$3.5 billion

Number of Annuitants:
~388,000

Industry Rank ('22 – Est.):
#4 Pension Risk Transfer Sales

Stable Value Investments

Core Pre-Tax GAAP Earnings:
~\$35 million (normalized)

AUM:
\$37 billion (SVIO)

Sales (SVIO):
'16A – \$0.8 billion
'17A – \$2.3 billion
'18A – \$3.5 billion
'19A – \$5.9 billion
'20A – \$7.2 billion
'21A – \$4.6 billion
'22A – \$7.6 billion

Industry Rank ('22 – Est.):
#1 SVIO provider (Flow)
#5 SVIO provider (AUM)

Funding Agreements

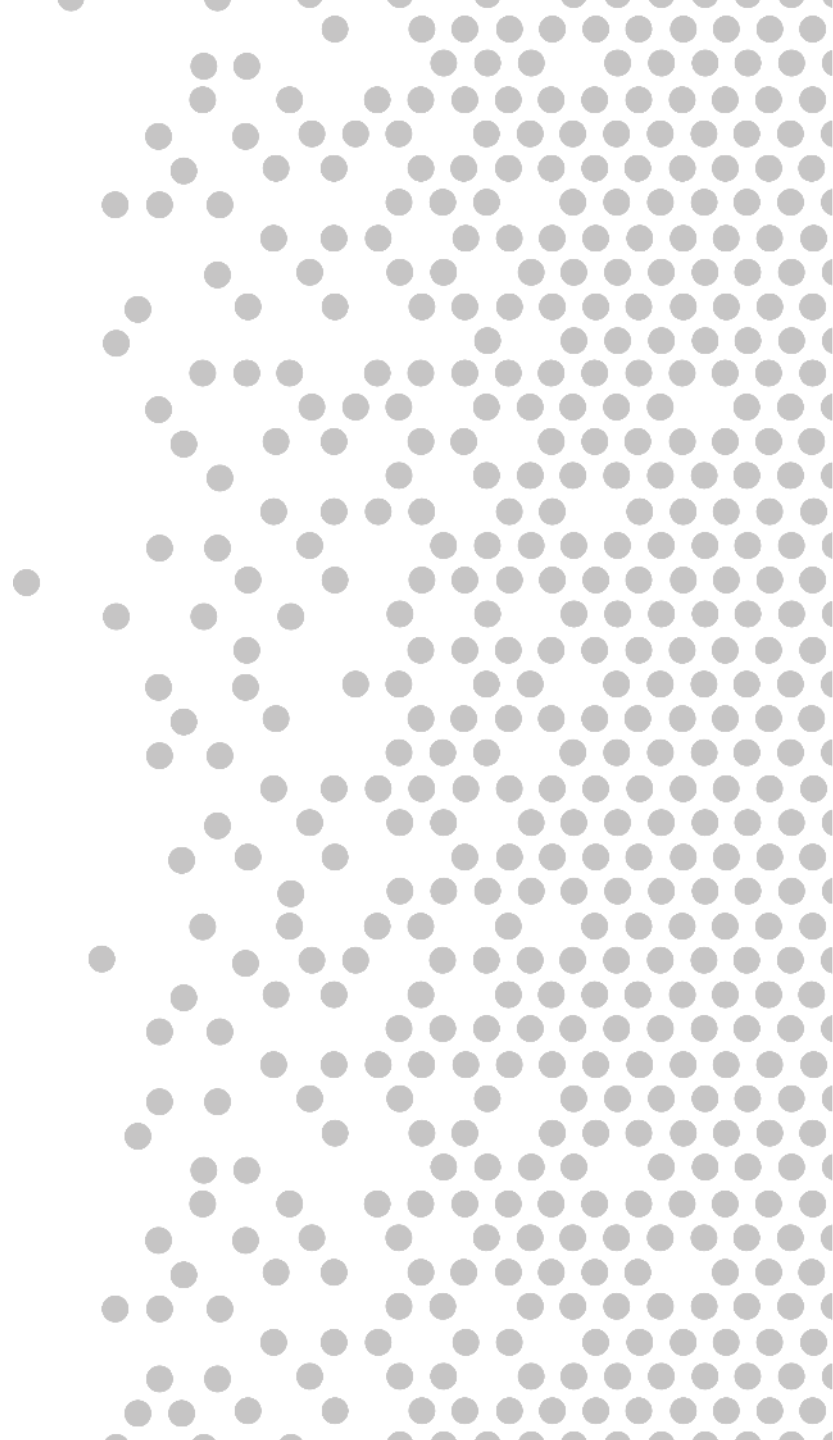
Core Pre-Tax GAAP Earnings:
~\$158 million

AUM:
~\$15 billion

Sales:
'16A – \$1.6 billion
'17A – \$2.0 billion
'18A – \$0.9 billion
'19A – \$2.5 billion
'20A – \$4.2 billion
'21A – \$4.0 billion
'22A – \$3.6 billion

Industry Rank ('22):
#7 FABN New Issuance

Strategic Focus & Financial Results



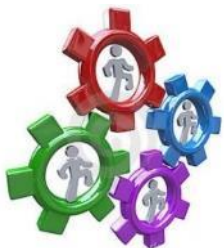
- Advance the profitable growth of Institutional Solutions in support of MM enterprise strategic goals
- Further extend MassMutual's market position, business relationships and long-term opportunities across the institutional markets
- Deliver strong risk-adjusted returns to contribute to earnings, dividends and capital
- Optimize the balance between profitability, business growth and risk management
- Generate top-line growth to increase the in-force book and add to MassMutual's customer base
- Continue to enhance and evolve the business capabilities and customer experience for institutions, intermediaries and individuals
- Invest in the underlying business infrastructure to support long-term sustainability



BUSINESS GROWTH

	<u>2015</u>		<u>2022</u>
Earnings*	\$351m		Earnings* \$874m
Sales	\$6.6b		Sales \$16.2b
AUM	\$47b		AUM \$105b

* Core Pre-Tax GAAP Earnings



MARKET FOCUS

<u>2015</u>		<u>Today</u>
Disparate Businesses within Product-Centric Organizations		Institutional Market Presence
		 Clear Business Identities
		 Customer Needs Focus
		 Growth Momentum



SOLUTIONS

2015

Monocentric Market Offerings

General Market Approach
(Opportunistic Go-to-Market)



Today

Expanded Offerings for Varying
Markets and Client Needs

Clear Market Identity
(Strategic Market Presence)



CUSTOMER EXPERIENCE

2015

Several years of deferred
business investment resulting in
dated customer experience

Manual and non-digital
processes with sub-optimal
service model in certain markets



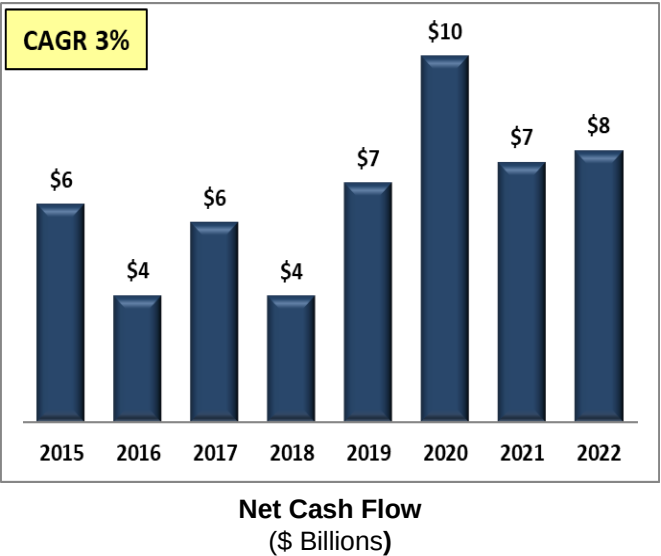
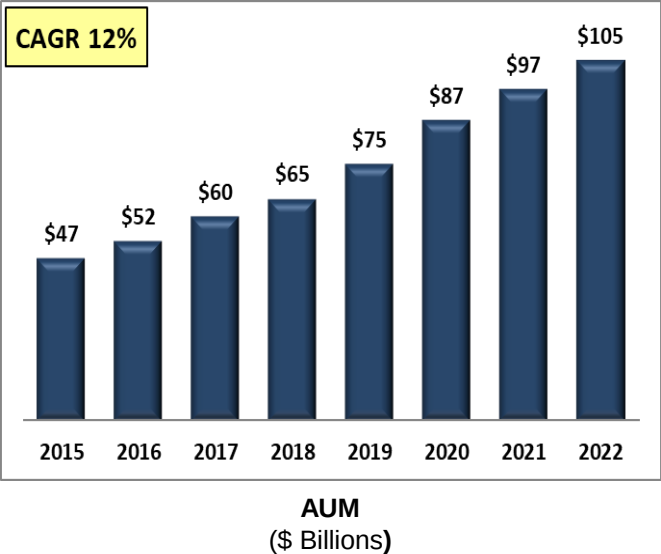
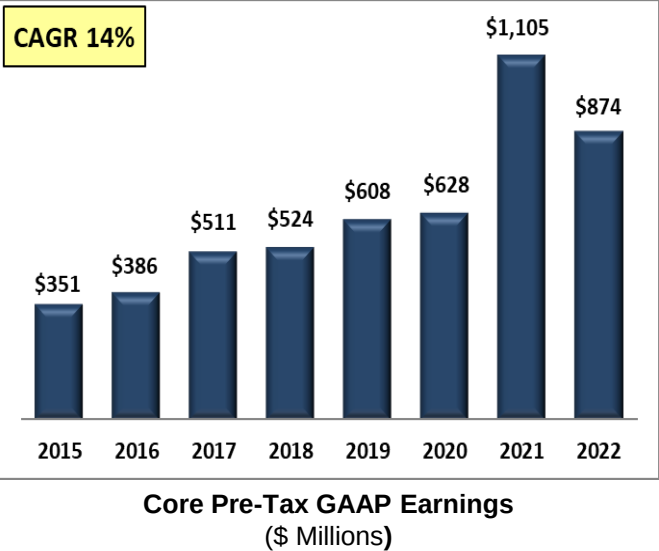
Today

Prioritized focus on service
models and capabilities, along
with digital and customer
experience investments

Enhancements include: web
portals, custom communications,
imaging, target marketing,
digital self-service, etc.

Institutional Solutions: Consistent Growth in In-Force Business and Profitability

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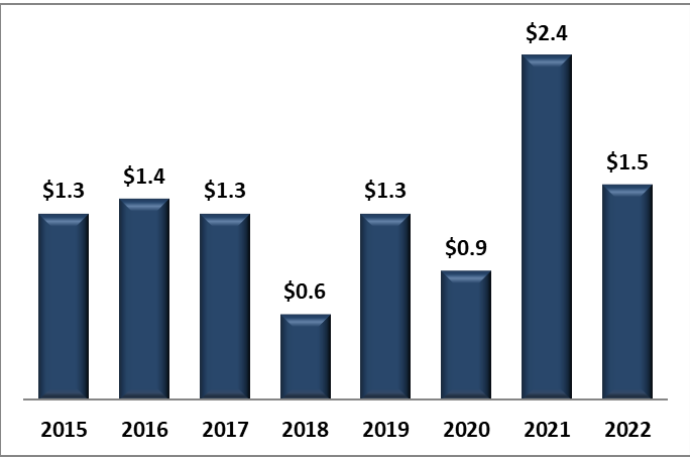
Institutional Solutions: Strong Sales Growth and Market Positions

January 2023

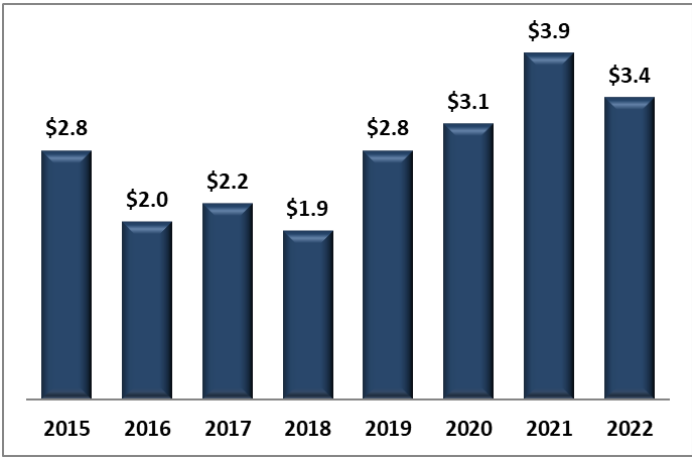
(\$ Billions)



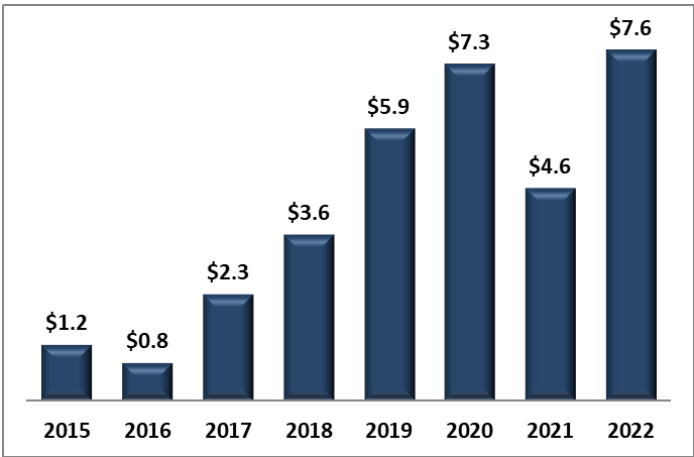
Institutional Solutions*



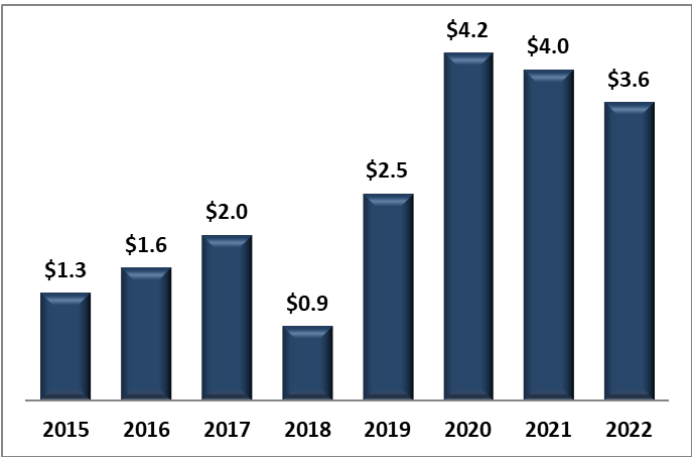
Institutional Insurance



Pension Risk Transfer



Stable Value Investments



Funding Agreements

Institutional Solutions

- IS Data Analytics Roadmap
- Salesforce Roadmap
- Digital Roadmap
- Marketing Presence
- Sub-Domain Update
- Thought Leadership

Institutional Insurance (BOLI/COLI)

- PPVUL iCOLI – SL21 Product Launch
- New Money Rate BOLI (SL19)
- SL20 PPVUL COLI Accredited Investors
- e-Delivery of Correspondence
- SL21 Statements using BEM process
- BOLI Automated Policy Assembly
- Digital Enhancements: Portal

Pension Risk Transfer

- PRT SIA Product Launch
- PRT Platform Replacement
- PRT Digital Retirement Initiation
- PRT Cashbook Automation
- Federal Tax Withholding
- Digital: Benefit Calcs, Portals, etc.

Stable Value Investments

- GIA IO Solutions
- CaRs & Cashbook Replacement
- CaRs & Cashbook Decommission
- CAS Decommission
- SV Underwriting Automation Tool
- Digital Enhancements

Funding Agreements

- CML Upgrade (SAP)
- Muni GIC Portal
- FABN NON-USD Currency
- Green FABN (ESG)
- Commercial Paper FABN

Addressing Business Challenges Remains at the Forefront of Focus

January 2023

- Ensuring the organization, technology and operational capabilities keep pace with multi-year business growth
- Extending enterprise capacity limits (e.g., capital, reinsurance, longevity, leverage, etc.) that impact growth potential
- Solving for investment-related yield and volume needs to support business growth, including access to constrained investment asset classes
- Enhancing adjacent product solutions that deepen relationships or address related customer needs
- Extending marketing and thought leadership in the institutional markets
- Expanding a competitive digital experience for institutional business customers
- Managing the talent needs (attraction, development, succession) for complex, high-growth businesses
- Navigating profitability and growth in a volatile and uncertain economic environment



- IS delivers significant economic value to MassMutual and to policyowners through scalable business growth and high earnings
 - Serves the financial needs of business customers via sustainable leadership positions across four distinct markets
 - Established leadership positions in the key institutional markets, with the goal to optimize profit, growth and risk
 - Value proposition focused on financial strength, expertise, innovation, flexibility, intermediary relationships and trust
- A focused approach to the institutional markets has enabled IS growth over the past several years and guides the business strategy into the future
 - External risks are factors that impact market opportunity such as economic conditions, interest risks, regulatory policy and non-traditional entrants
 - Internal factors influence growth potential including enterprise capital constraints, reinsurance limits, risk capacity (e.g., longevity) and infrastructure
 - Pursuit of attractive opportunities and solutions to business challenges remain at the forefront of focus
- IS strategy delivers favorable return businesses that provide capital-efficient and early-emergent profitability supporting earnings growth, policyowner dividends, statutory profits, capital generation and enterprise strategic investments

